



EAST WEST UNIVERSITY

Pre-Lab

Experiment No: 3

Experiment Title: Behavioral Verilog Simulation of a Combinational Logic Circuit

Course Code: CSE-345

Section: 03

Submitted by

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Objective:

1. To learn behavioral Verilog coding of a combinational logic circuits using procedural model.
2. To learn behavioral Verilog coding of a combinational logic circuits using continuous assign statement.

Equipment:

Quartus II software

Circuit Diagram:

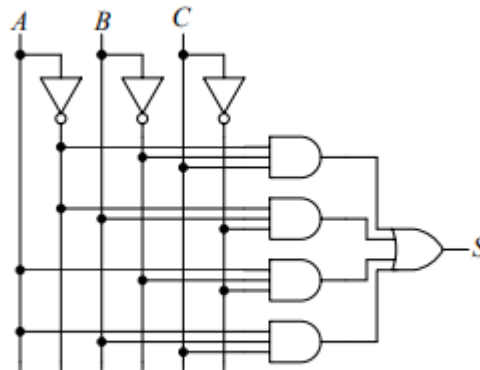


Figure 2: Combinational circuit for lab practice.

Procedure:

1. Simulate both Verilog codes prepared in the pre-lab report using Quartus II software.
2. Obtain printouts of both Verilog codes along with their respective simulation waveforms.
3. Arrange the Verilog codes and simulation results of both cases on a single page.
4. Submit the printed page to the instructor for verification and signature

Pre-Lab Question Answers:

1. Write behavioral Verilog code for the combinational logic circuit of Figure 2 using procedural model.

Ans:

From the circuit in Figure 2, the four AND gates produce the following terms:

- Gate 1 = $A'B'C$
- Gate 2 = $A'BC'$
- Gate 3 = $AB'C'$
- Gate 4 = ABC

Therefore,

$$S = A'B'C + A'BC' + AB'C' + ABC$$

The behavioral Verilog code using the procedural model is given below: module

lab3_1 (input A, B, C,

output reg S);

always @(A, B, C) begin

S = 0; if($\sim A$ &
 $\sim B$ & C)

S = 1;

if($\sim A$ & B & $\sim C$)

S = 1;

if(A & $\sim B$ & $\sim C$)

S = 1;

if(A & B & C)

S = 1;

end endmodule

2. Write behavioral Verilog code for the combinational logic circuit of Figure 2 using continuous assign statement.

Ans:

Using the same Boolean expression, the continuous assign Verilog code is: module

```
lab3_2 (input A, B, C,  
        output S);  
    assign S = (~A & ~B & C) |  
              (~A & B & ~C) |  
              (A & ~B & ~C) |  
              (A & B & C);  
endmodule
```